Today's topics

1. Joint probability distribution
2. Marginal probability distribution
3. Covariance of two random variables
4. Properties of covariance
5. Correlation coefficient
6. GATE-level MCQ, MSQ and NAT practice

Topic 1: Understanding two random variables

Until now, we studied one random variable X.

Now imagine two random variables:

- X = Number of hours studied.
- Y = Marks obtained in an exam.

We want to understand how X and Y behave together.

Topic 2: Joint probability distribution

Joint probability means the probability of two events or random variables taking particular values together.

$$P(X = x, Y = y)$$

Topic 3: Marginal probability distribution

Marginal probability means finding the probability distribution of X alone or Y alone from the joint distribution.

Example

Suppose two fair coins are tossed.

Let:

- X = 1 if the first coin is Head, otherwise 0.
- Y = 1 if the second coin is Head, otherwise 0.

Possible outcomes:

Outcome	X	Y	Probability
HH	1	1	1/4
HT	1	0	1/4
TH	0	1	1/4
TT	0	0	1/4

Joint probability table:

X \ Y	0	1
0	1/4	1/4
1	1/4	1/4

For X:

$$P(X = 0) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

$$P(X = 1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

For Y:

$$P(Y = 0) = \frac{1}{2}, \quad P(Y = 1) = \frac{1}{2}$$

Joint and marginal distribution – Mini test

3 MCQs

Q1. What does P(X=1,Y=0) represent?

- A. Probability that X=1 only
- B. Probability that Y=0 only
- C. Probability that X=1 and Y=0 together
- D. P(X=1)+P(Y=0)

Q2. In the two-coin example, find P(X=1).

- A. 1/4
- B. 1/2
- C. 3/4
- D. 1

Q3. To obtain the marginal distribution of X from a joint table, we:

- A. Multiply all probabilities
- B. Add probabilities across each X row
- C. Subtract columns
- D. Square probabilities

Topic 4: Covariance

1. What is covariance?

Covariance measures how two random variables change together.

For example:

- X = Hours studied.
- Y = Marks obtained.

If study hours increase and marks also tend to increase, covariance is positive.

2. Covariance formula

$$\boxed{Cov(X,Y) = E[XY] - E[X]E[Y]}$$

3. Understand using an example

Consider two fair coins.

X = 1 if the first coin is Head, otherwise 0.

Y = 1 if the second coin is Head, otherwise 0.

The possible outcomes are:

Outcome	X	Y	XY
HH	1	1	1
HT	1	0	0
TH	0	1	0
TT	0	0	0

Calculate:

$$E[X] = \frac{1}{2}$$

$$E[Y] = \frac{1}{2}$$

$$E[XY] = \frac{1}{4}$$

$$Cov(X,Y) = \frac{1}{4} - \frac{1}{2} \times \frac{1}{2}$$

$$= \frac{1}{4} - \frac{1}{4}$$

$$\boxed{Cov(X,Y) = 0}$$

4. Properties of covariance

Remember these formulas:

Property 1:

$$Cov(X,X) = Var(X)$$

Property 2:

$$Cov(X,Y) = Cov(Y,X)$$

Property 3:

$$Cov(X,c) = 0$$

Property 4:

$$Cov(aX,bY) = abCov(X,Y)$$

Property 5:

$$Cov(X+Y,Z) = Cov(X,Z) + Cov(Y,Z)$$

Important: If X and Y are independent, then:

$$Cov(X,Y) = 0$$

Q1. If E[X]=3, E[Y]=4 and E[XY]=15, find Cov(X,Y).

- A. 3
- B. 12
- C. 15
- D. 27

Q2. Cov(X,X) is equal to:

- A. E[X]
- B. Var(X)
- C. $E[X^2]$
- D. 0 always

Q3. If X and Y are independent, then:

- A. Cov(X,Y)=1
- B. Cov(X,Y)=0
- C. Cov(X,Y)=Var(X)
- D. Cov(X,Y) is always positive

Q4. If Cov(X,Y)=5, find Cov(2X,3Y).

- A. 10
- B. 15
- C. 25
- D. 30

Correlation coefficient

1. What is correlation?

Correlation measures the **strength and direction of the linear relationship** between two random variables.

2. Formula

$$\boxed{\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y}}$$

Or,

$$\rho_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$$

3. Important range

$$\boxed{-1 \le \rho_{XY} \le 1}$$

Understanding correlation values

+1 **Perfect positive correlation**

X increases → Y increases linearly.

0 **Zero linear correlation**

No linear relationship. Nonlinear relationship may still exist.

-1 **Perfect negative correlation**

X increases → Y decreases linearly.

4. Solved example

Given:

- Cov(X,Y) = 6
- Var(X) = 4
- Var(Y) = 9

Find correlation.

$$\rho = \frac{6}{\sqrt{4 \times 9}}$$

$$= \frac{6}{\sqrt{36}} = \frac{6}{6}$$

$$\boxed{\rho = 1}$$

Answer: Perfect positive correlation.

5. Covariance vs Correlation

Covariance	Correlation
Unbounded	Between -1 and +1
Depends on units	Unitless
Measures direction and joint variation	Measures direction and strength of linear relationship
Cov(X,Y)	Cov(X,Y)/(σXσY)

GATE important points:

- Independence implies zero covariance (if moments exist).
- Zero covariance does not necessarily imply independence.
- Zero correlation does not necessarily imply independence.
- Adding a constant to a variable does not change correlation.
- Multiplying one variable by a negative constant reverses the sign of correlation.

Q1. If Cov(X,Y)=12, Var(X)=16 and Var(Y)=9, find ρ.

- A. 0.5
- B. 1
- C. 2
- D. 0.25

Q2. The value of correlation coefficient always lies in:

- A. 0 to 1
- B. $-\infty$ to $+\infty$
- C. -1 to $+1$
- D. 0 to $+\infty$

Q3. If Corr(X,Y)=0, which statement is always true?

- A. X and Y are independent
- B. There is no linear correlation
- C. There is no relationship of any kind
- D. Cov(X,Y)=1

Q4. If Corr(X,Y)=0.8, what is Corr(X+5,Y)?

- A. 0.8
- B. 5.8
- C. −0.8
- D. 0

Section A – MCQ

Q1. The joint PMF of two random variables X and Y is given below.

X \ Y	0	1
0	0.2	0.3
1	0.1	0.4

Find $P(X = 1 \mid Y = 1)$.

A. 0.4

B. 0.5

C. 0.5714

D. 0.8

Q2. Given:

$$E[X] = 2, \quad E[Y] = 3, \quad E[XY] = 8$$

Find $\mathrm{Cov}(X, Y)$.

A. 2

B. 6

C. 8

D. 14

Q3. If $\mathrm{Var}(X) = 4$, $\mathrm{Var}(Y) = 9$, and $\mathrm{Cov}(X, Y) = -3$, find the correlation coefficient.

A. −1

B. −0.5

C. 0.5

D. 1

Section B – MSQ

Select all correct options. Partial selection is not counted as fully correct.

Q4. Which statements are always true, assuming the required moments exist?

A. If X and Y are independent, then Cov(X,Y) = 0.

B. If Cov(X,Y) = 0, then X and Y are independent.

C. Cov(X,Y) = Cov(Y,X).

D. Cov(X,X) = Var(X).

Q5. Which statements about correlation are correct?

A. Correlation is always between −1 and +1.

B. Adding a constant to X changes Corr(X,Y).

C. Multiplying X by −2 reverses the sign of Corr(X,Y).

D. Zero correlation always implies independence.

Q6. A joint PMF is:

$$p(x,y) = \frac{x+y}{12}$$

for $x \in \{0,1,2\}$, $y \in \{0,1\}$.

Which statements are correct?

A. $P(X=0) = 1/4$

B. $P(X=1) = 1/3$

C. $P(Y=1) = 1/2$

D. $P(X=2) = 5/12$

Section C – NAT

Q7. Given the following joint distribution:

X \ Y	0	1
0	0.1	0.2
1	0.3	0.4

Calculate $E[X]$.

Q8. For the same joint distribution in Q7, calculate $E[Y]$.

Q9. For the same distribution, calculate:

$$\mathrm{Cov}(X, Y)$$

Q10. Given:

$$E[X] = 4, \quad E[Y] = 5$$

$$\mathrm{Var}(X) = 4, \quad \mathrm{Var}(Y) = 25$$

$$\mathrm{Cov}(X, Y) = 6$$

Calculate the correlation coefficient.